Predicting Late ETA Risk

Upper-bound ETA miss probability (binary classification)

Elnur Garibov

Email: elnur.garibov@aalto.fi

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Goal

Predict P(late) where late = 1 if actual_delivery_time_minutes > estimated_delivery_time_upper_minutes.

Dataset: simulated Helsinki orders (Spring 2022).

Problem and decision context

- Problem: predict whether an order will arrive later than the promised upper ETA bound.
- Who uses it: ETA/customer experience teams and Operations (dispatch, supply management).
- Decision supported: widen ETA range or trigger proactive 'delay risk' messaging for at-risk orders.
- Why relevant: repeated upper-bound misses hurt customer trust, NPS, and increase support contacts.

Target definition

late = 1 if actual_delivery_time_minutes > estimated_delivery_time_upper_minutes

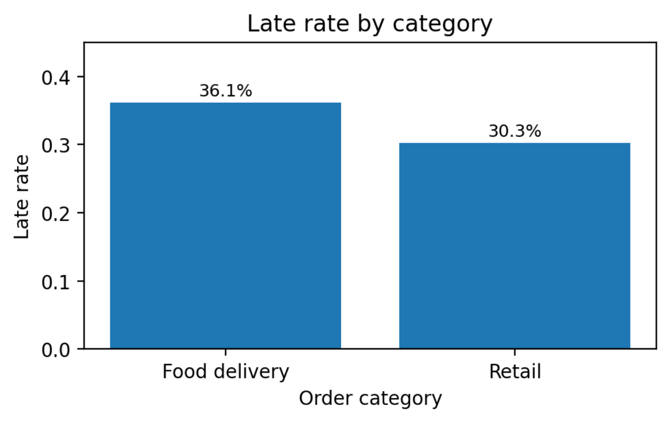
Model output: probability of late risk used for threshold-based actions.

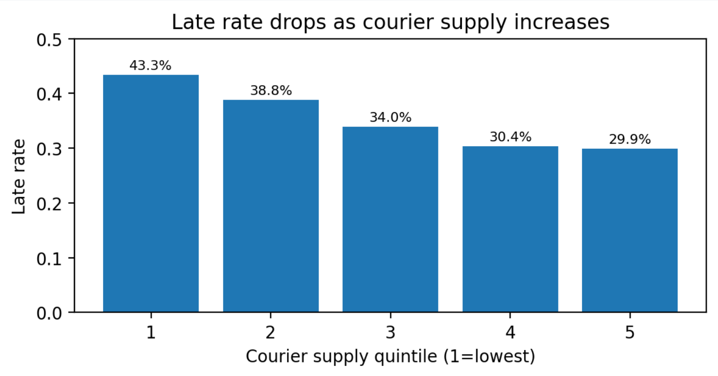
Data and EDA findings

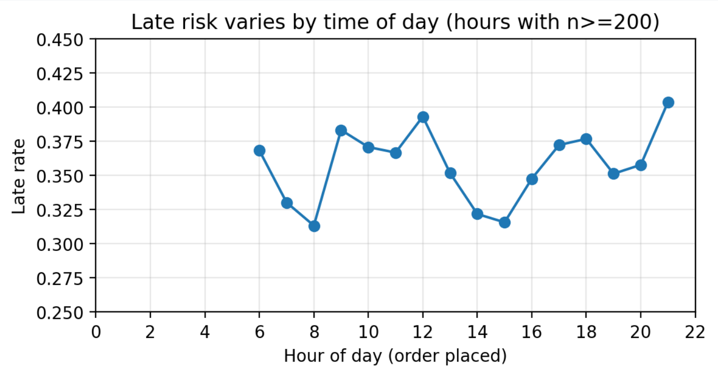
Total orders	Train / Test split	Overall late rate	Test late rate
24,942	**80% / 20%**	**35.4%**	**34.2%**

Key EDA insights (actionable)

- Category signal: Food delivery late-rate 36.1%, Retail 30.3%.
- Supply matters: lowest supply quintile late-rate 43.3% vs highest 29.9%.
- Time-of-day: higher late risk around 09-12 and 17-18 (commute/meal peaks).







Feature engineering and representations

Signals used

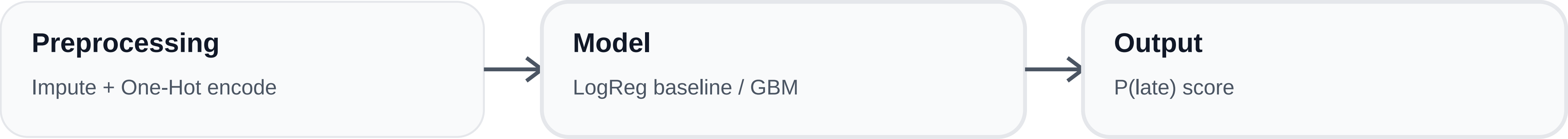
- Numeric: item_count, courier_supply_index, precipitation
- ETA bounds: eta_lower, eta_upper (upstream system output)
- Time: hour of day, day-of-week, weekend flag
- Geo proxy: same_h3 (venue and customer in same H3 cell)
- Coarse region: venue_reg = first 5 chars of venue H3
- Category: order_category

Signals intentionally not used

- Full H3 as category: high-cardinality -> overfitting and heavier model
- Exact H3 distance / routing time: likely useful but skipped for simplicity
- Raw timestamp: replaced by hour/dow/weekend to reduce drift sensitivity
- Post-order / live signals: avoided to prevent leakage

Design choice: simple, low-leakage signals available at order-placed time.

Modeling approach and assumptions



Choices

- Task: binary classification (late vs on-time).
- Baseline: Logistic Regression (fast, interpretable).
- Non-linear: Gradient Boosting (captures interactions).
- Validation: chronological split to mimic production.

Key assumptions

- Relationship between inputs and late-risk is reasonably stable over time.
- ETA bounds are generated consistently by upstream system (changes require recalibration).
- Coarse geo features are sufficient for a first-pass model.

Results and evaluation

Test-set metrics (probability + ranking)

Model	ROC-AUC	PR-AUC	Brier (lower=better)
Logistic Regression	0.633	0.501	0.215
Gradient Boosting	0.643	0.506	0.211

Operational threshold example (thr = 0.30)

Test set: 4,989 orders (on-time=3,282, late=1,707)

Model	Late recall	Late precision	On-time recall	Accuracy
LogReg	0.869	0.363	0.208	0.434
GBM	0.755	0.402	0.417	0.532

What the numbers mean

- thr=0.30 prioritizes catching late orders (recall) over raw accuracy.
- False negatives are worst: optimistic ETA -> disappointment.
- False positives are tolerable: conservative ETA/warning improves reliability.

Limitations

Known limitations

- Simplified geo representation: no explicit distance, routing, or travel-time features.
- Limited external context: no traffic, events, or detailed weather severity.
- Binary target ignores severity (how late) and focuses only on missing upper bound.

When this breaks in production

- Drift from upstream ETA changes or shifts in courier supply dynamics (seasonality, major events).
- New regions/venues or changing order mix reduce reliability of coarse region features.
- Probability calibration can degrade without monitoring and periodic retraining.

Next steps

If deployed, I would do next:

- Feature upgrades: H3 distance/travel-time proxies; region-hour historical late-risk (congestion patterns).
- Probability calibration (Platt or isotonic) so thresholds map cleanly to business actions.
- Segmented thresholds (category/region/hour) with cost-aware evaluation aligned to customer trust.
- Monitoring: late-rate drift, calibration drift, feature distribution shift; alerting and retrain triggers.
- Retraining cadence (e.g., weekly) with evaluation dashboards.

Goal: maximize ETA reliability and customer trust using simple, low-leakage signals.